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Perspective independence, more than personas, drives LLM teams — and where they reverse

Running title: When does a team of LLMs beat one call?

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Abstract

Multi-agent prompting of large language models has produced contradictory diagnostic results, and it remains unclear whether any benefit comes from specialist personas or from perspective independence. We compared a single direct call, five personas in one context, and the same five roles as isolated agents integrated by a moderator, with five repeat runs per case, on 87 CPC cases, 406 MedCaseReasoning cases, and 364 emergency department encounters, under an LLM judge validated against clinicians. On the external benchmark the team beat the single call on both pre-specified recall endpoints (top-3 +3.0 points, $p = 0.0079$; top-5 +3.9, $p = 3.8 \times 10^{-5}$); a factorial attributes the gain to independent generation plus moderated synthesis, not the specialist roles. On real emergency presentations the benefit reversed (top-1 40.1% versus 34.3%, $p < 0.0001$), carried by the specialist role lists and surviving added objective results. Deployment should key on the question and the input at hand.

Keywords: Large Language Models; Multi-Agent Systems; Diagnostic Reasoning; Diagnosis, Computer-Assisted; Differential Diagnosis; Artificial Intelligence; Emergency Medicine

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Introduction

Diagnostic error remains one of the largest unaddressed sources of patient harm: most people will experience at least one diagnostic error in their lifetime [1], and a national analysis attributed roughly 795,000 deaths or permanent disabilities each year in the United States to misdiagnosis [2]. Large language models (LLMs) have progressed from encoding clinical knowledge [3] to performing competitively on benchmarks built for expert diagnosticians — clinicopathological conference (CPC) cases [4,5], randomized comparisons against primary care physicians [6,7]. What remains unsettled is not whether these models can reason about diagnosis, but how that reasoning should be organized.

A popular answer is to simulate a team: specialist personas, conversational multi-agent frameworks, or aggregation of independent generations by voting [8,9,10,11], on the intuition, borrowed from multidisciplinary team practice [12], that several perspectives improve diagnosis. The evidence is contradictory. Multi-agent conversational frameworks report substantial gains on rare-disease cases [8,13], yet system-prompt personas produce no stable improvement on objective tasks [14], debate can entrench error without external feedback [15], automatically constructed multi-agent systems underperform budget-matched chain-of-thought baselines [16,17], and large-scale syntheses find the benefit of collaboration shrinks as backbone capability grows [22], vanishes under equalized thinking-token budgets [23], and eludes tool-using clinical agents [24]. Positive reports rarely include budget-matched single-model controls, so accuracy gains remain entangled with test-time compute.

One reason is that two distinct variables have been conflated. “Multiple perspectives” can mean several personas voiced sequentially within one context, where each perspective conditions on the ones before it, or genuinely independent opinions generated in isolation and only then combined. No prior work has isolated independence as the sole manipulated variable while holding roles, model, and prompt content constant. A second gap is distributional: multi-agent diagnostic prompting has been evaluated almost exclusively on curated, information-rich case reports, not on real emergency department presentations, where the first-contact input is thin and the reference diagnosis is often the treating clinician’s working label rather than a confirmed disease. Thin input is the natural explanation for a benefit that disappears there and, unlike the first confound, it is testable within a single dataset by adding the visits’ objective results to the input.

Here we address both gaps. In a single large language model we compare a single-pass direct call, five personas voiced within one context, and the same five roles run as isolated, mutually blinded agents integrated by a moderator; the second and third differ only in whether perspectives are independent, so the gap between them is attributable to isolation alone, and a budget-matched control of five independent samples with rank aggregation further separates structure from test-time compute. We evaluate on three datasets differing in difficulty, provenance, and reference-label structure — 87 CPC cases, 406 from an external reasoning benchmark, 364 real emergency department encounters — with a semantic-equivalence LLM judge validated against blinded clinicians on every corpus [19]. We find the benefit of multi-perspective reasoning real but conditional: produced by independence of generation together with moderated synthesis, not personas; concentrated in differential-list recall; and reversed on real emergency presentations, where a single direct call is best — a reversal that survives adding the visits’ objective results and is carried by the specialist role lists rather than multi-

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call inference itself. These findings convert a contested literature into deployment guidance keyed to the clinical question and the input at hand.

Results

The LLM judge reaches human-level agreement

Before any strategy comparison, we validated the semantic-equivalence judge against two clinicians in a double-blind study of 100 candidate–reference pairs that no rule or prompt had touched during development: GLM agreed with the raters at $\kappa = 0.814$ and 0.878 against an inter-rater level of 0.812 — with 100 pairs, agreement at, not above, the inter-rater level; the validation was then extended to the two non-CPC corpora with the same protocol (100 judge-stratified pairs per corpus, half boundary cases), where on the emergency set's symptom-level labels the criterion itself is softer — raters agree with each other at $\kappa = 0.612$ (0.81 on case-report labels), and the judge tracks that level (κ 0.640 and 0.580) (Supplementary Tables S5, S5-2). The judge and rules were frozen before the outcome analyses; because every arm in a contrast is scored by the same judge, its known leniency asymmetry cancels in paired comparisons, and on ER-Reason — where the multi-perspective arms produce longer, more etiologically elaborate candidates — leniency works against, not for, the reversal we report. Every main result is additionally reported under a second judge from a different model family and under a judge-independent word-coverage check (Supplementary Note S18).

Isolation turns the persona penalty into a recall advantage

On the 87 CPC cases, voicing five personas within a single context (P) did not help: P scored below the single direct call (A×1) on every metric, a directional trend under the case-level tests (top-1, $p = 0.24$; top-5, $p = 0.12$). Isolating the same five roles and integrating them through a moderator (MDT) reversed the picture: MDT exceeded A×1 on both pre-specified primary endpoints (top-3: +5.5 percentage points, 95% CI 1.1 to 10.3, Wilcoxon $p = 0.0079$; top-5: +4.4 points, 95% CI 0.9 to 8.3, $p = 0.015$) and exceeded P on both recall endpoints (top-3, +9.4 points, $p = 0.0043$; top-5, +8.7 points, $p = 0.0036$) (Table 1, Figures 1 and 2). These estimates include the 41 prompt-development cases and are correspondingly optimistic (the development subset carried roughly twice the held-out margin); the held-out subset (n = 46) is a direction check, and the significance claim rests on the independent external replication.

Table 1. Diagnostic performance of the three strategies across datasets (primary judge, GLM × v3).

Dataset	Strategy	Top-1 (%)	Top-3 (%)	Top-5 (%)
CPC (n = 87)	A×1	57.9 ± 1.9	76.1 ± 4.1	81.1 ± 2.6
	P	53.8 ± 4.0	72.2 ± 2.6	76.8 ± 2.1
	Pfixed (prompt-corrected)	52.4 ± 1.3	75.6 ± 3.0	80.9 ± 1.9
	MDT	60.5 ± 3.1	81.6 ± 2.4	85.5 ± 2.1
MCR (n = 406)	A×1	57.0 ± 0.9	69.9 ± 0.6	75.6 ± 0.8

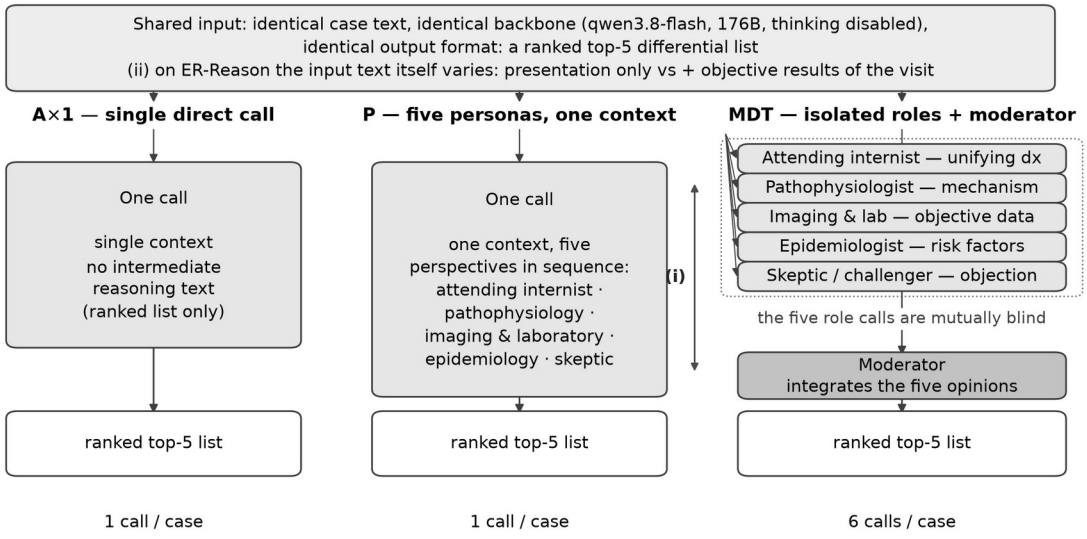
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Dataset	Strategy	Top-1 (%)	Top-3 (%)	Top-5 (%)
	P	56.4 ± 0.3	69.2 ± 1.1	75.0 ± 1.0
	MDT	55.9 ± 0.8	72.9 ± 0.8	79.6 ± 0.8
ER-Reason (n = 364)	A×1	40.1 ± 0.9	54.6 ± 1.1	61.6 ± 0.8
	P	37.7 ± 0.6	48.5 ± 1.5	55.2 ± 0.8
	MDT	34.3 ± 1.0	45.5 ± 1.0	51.9 ± 1.1

All values are mean ± SD across five seeds; hypothesis tests use per-case hit rates across seeds (paired Wilcoxon; Statistical analysis). Full per-seed values and sensitivity-judge versions of all figures appear in Supplementary Tables S1 and S2; legacy pooled-seed McNemar figures in Supplementary Table S3. The second model family and the additional control arms (reasoning elicitation, matched call budget, prompt-corrected P, and backbone scale) are summarized in Figure 5 and Supplementary Tables S10–S12. Temperature-matched control (all arms at 0.3) leaves every conclusion unchanged (Supplementary Note: Temperature-matched analyses). The P arm carried a prompt transcription defect; the Pfixed row is its corrected control (Methods).

Figure 1. Study design: three prompting strategies and the two controlled contrasts.



(i) P vs MDT, the controlled contrast: roles, prompt material, model and output format are held fixed; the only change is whether the five perspectives are isolated from one another.
(ii) ER-Reason input contrast: the same 364 encounters, identical prompts, model, decoding, seeds and judge; the only change is the input text — presentation only (ED provider note truncated before "Medical Decision Making") versus presentation + objective results of the same visit (laboratory values, vital signs, imaging / ECG / ultrasound reports).
Endpoint: the reference diagnosis appears within the top-1, top-3 or top-5 candidates, adjudicated by GLM-5.3-flash (v3 rules, blinded to strategy). Call counts per case (token_costs.json): A×1 = 1, P = 1, MDT = 6.

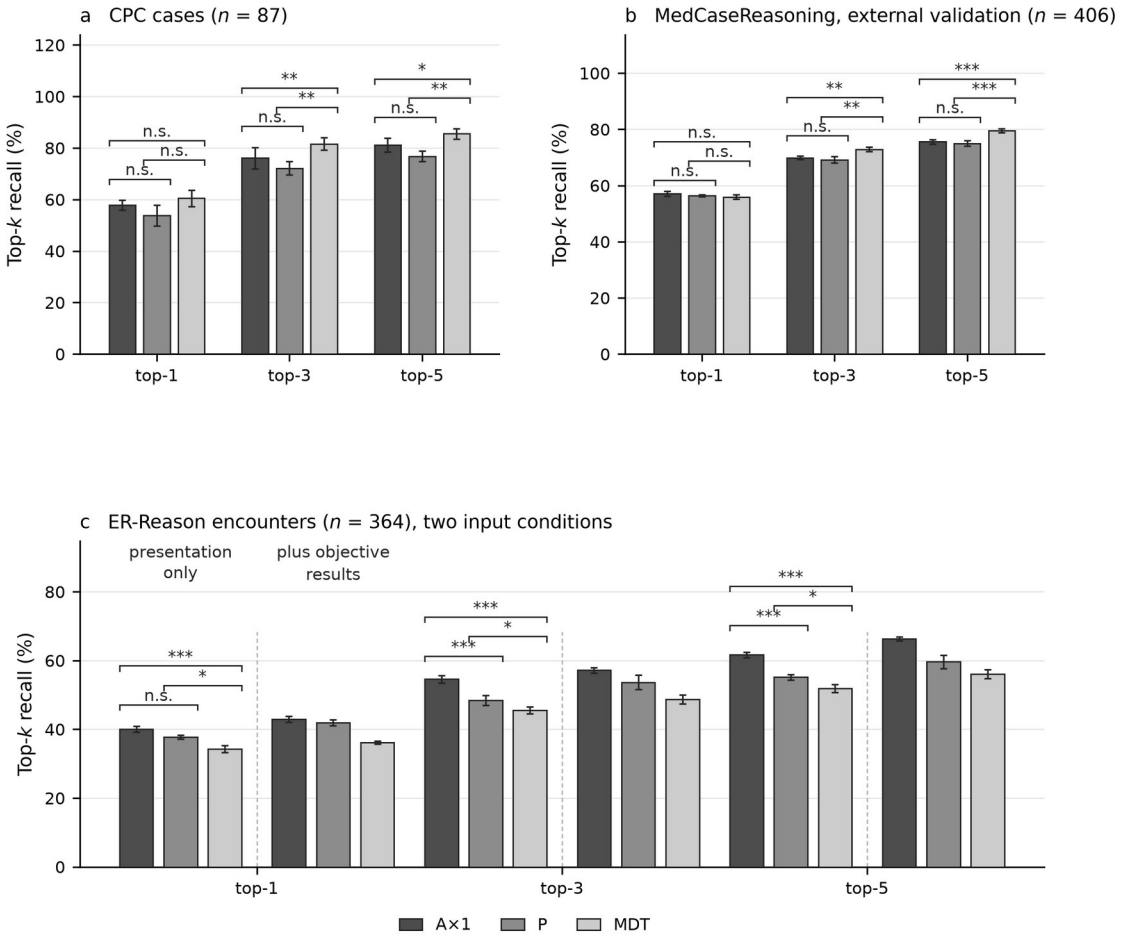
Three strategies were run on the same backbone (qwen3.8-flash, 176B parameters, thinking mode disabled) and returned the same output format, a ranked list of five differential diagnoses. **A×1** is a single zero-shot call that emits the ranked list directly, with no intermediate reasoning text (1 call per case). **P** reasons from the same five perspectives inside one shared context, so each perspective conditions on the previous ones (1 call per case). **MDT**

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issues those five roles as independent calls that cannot see one another's output, and a moderator call then integrates the five opinions into the final list (5 role calls + 1 moderator call = 6 calls per case). Call counts are from the token accounting (Table 3). Contrast **(i)** holds roles, prompt material, model, decoding settings and output format fixed, manipulating only whether the five perspectives are isolated from one another — the sole difference between P and MDT. Contrast **(ii)** is the ER-Reason input manipulation: the same 364 encounters, prompts, model, decoding, seeds and judge are run under two input texts — presentation only (the provider note truncated immediately before the "Medical Decision Making" section) and presentation plus the visit's objective results (laboratory values, vital signs, and imaging reports appended under a short heading). Performance was scored by whether the reference diagnosis appears within the top-1, top-3 or top-5 candidates (GLM-5.3-flash, frozen v3 rules, blinded to strategy).

Figure 2. Top-k recall of the three strategies on (a) CPC, (b) MedCaseReasoning and (c) ER-Reason under five seeds, with the ER-Reason input contrast.



Primary judge: GLM-5.3-flash x v3. Error bars: +/-1 SD across the five seeds. Brackets: paired Wilcoxon signed-rank test on per-case five-seed hit rates (*** p < 0.001, ** p < 0.01, * p < 0.05, n.s. = not significant).

Bars give mean top-1, top-3 and top-5 recall over five independent seeds; error bars ± 1 SD (sample SD, ddof = 1). Panels a and b: CPC (n = 87) and the external MedCaseReasoning set (n = 406). Panel c: the same 364 emergency department encounters run under two input

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conditions; within each endpoint the left triplet of bars is the presentation-only input and the right triplet is the same cases with the visit's objective results added, separated by a dashed rule. Brackets show the paired two-sided Wilcoxon signed-rank test on per-case five-seed hit rates (the case is the unit of inference); *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant; all p values are unadjusted, and 15 of the 16 primary-framework contrasts with $p < 0.05$ survive Benjamini–Hochberg correction (Supplementary Table S4). All values are tabulated in Tables 1 and 2 and Supplementary Tables S1 and S6; the strategy-by-input interaction in panel c is null at every endpoint (all $p \geq 0.148$, Supplementary Table S6), and the MedCaseReasoning MDT runs carry the data-integrity repair described in Methods.

One further control excludes backbone scale: 176B ⟶ 2400B raised the single-call strategies with no scale-by-strategy interaction (all $p \geq 0.43$), and the six-call 176B team was indistinguishable from a single 2400B call ($p = 0.90$; Supplementary Table S12).

External replication and the completed factorial

External validation on 406 MedCaseReasoning cases — an independent public benchmark where the MDT-versus-A×1 contrast is significant at both recall endpoints — replicated the pattern at scale: top-1 was statistically indistinguishable (55.9 ± 0.8% versus 57.0 ± 0.9%, $p = 0.25$), while MDT was superior on top-3 (72.9 ± 0.8% versus 69.9 ± 0.6%, +3.0 points, 95% CI 0.9 to 5.1, $p = 0.0079$) and top-5 (79.6 ± 0.8% versus 75.6 ± 0.8%, +3.9 points, 95% CI 2.1 to 5.9, $p = 3.8 \times 10^{-5}$), and beat P on both recall endpoints ($p = 0.0027$ and $p = 1.0 \times 10^{-4}$). On the 46 held-out CPC cases every MDT-versus-A×1 contrast matched the full set without reaching significance (top-3 $p = 0.096$ on 12 non-tied cases) — the power of a 46-case set, not inconsistency — so the recall advantage stands on this external replication (Supplementary Table S7; Supplementary Note S20).

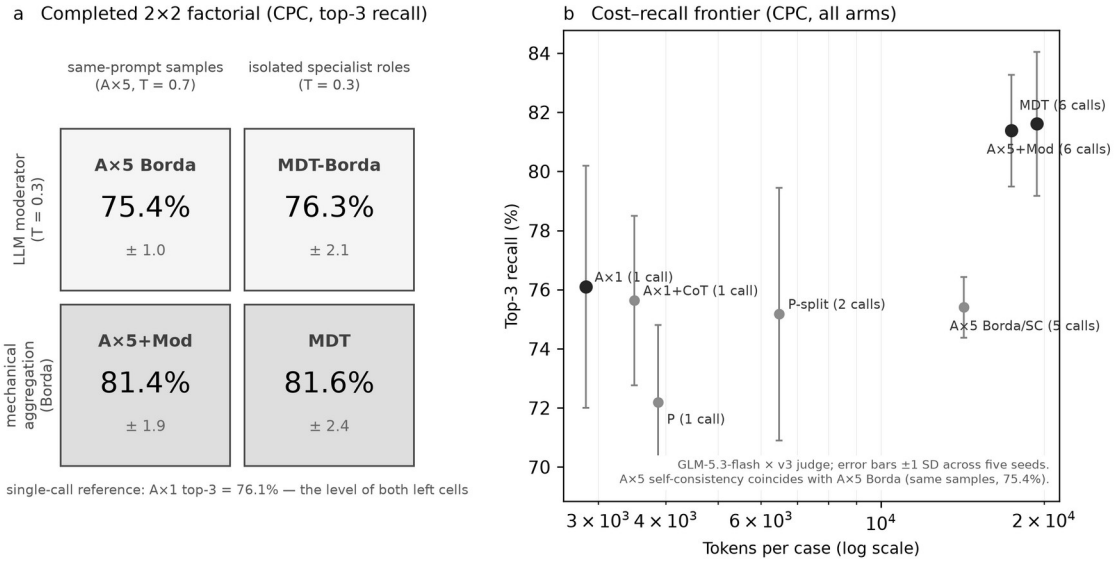
The completed factorial then pins the attribution (Figure 3a). Mechanical aggregation of the lists never helped, from either source: five independent samples of the single-call prompt aggregated by Borda count scored 47.5 ± 2.1% at top-1 on MedCaseReasoning, 9.5 points below the single call, and re-aggregating the team's own five role lists by the same Borda rule (MDT-Borda) left them at or below the single call on CPC (indistinguishable at top-3 and top-5, −5.5 points at top-1, $p = 0.086$; −8.0 below the full team at top-1, $p = 0.0044$). Moderated synthesis, by contrast, recovered the team's advantage from either source: the same moderator over the same plain samples (A×5+Mod) added +10.6 points at top-1 over Borda on MedCaseReasoning and +9.0 on ER-Reason, and matched or beat the role-based team at every endpoint on both corpora (top-5 within 0.5 points of MDT, $p = 0.52$ on MedCaseReasoning). Neither the reasoning text nor the call budget explains the gain: eliciting a step-by-step justification inside the single call (A×1+CoT) moved no endpoint beyond noise (all $p \geq 0.61$), and the five-call budget spent on independent sampling with self-consistency aggregation likewise reproduced the single call, not the team (all contrasts against MDT $p \leq 0.059$; Supplementary Note S21, S22). A fifth cell locates the persona failure at generation time: when a fresh moderator integrates five sequentially generated, mutually conditioning perspective analyses (P-split), the result stays with P at top-1 (−6.5 points below the single call) even though the independent adjudicator recovers 6.4 points of top-5 recall over P ($p = 0.0034$) — once the voices have conditioned on one another, the candidate pool has already narrowed (Supplementary Note S24). Across all three corpora the moderator-over-mechanical contrast reads +7.6 (CPC), +9.0 (ER-Reason), and +10.6 (MedCaseReasoning) points at top-1, with the specialist framing of the

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lists optional everywhere (Supplementary Table S10i), under one decoding caveat — plain samples at 0.7 and role lists at 0.3, so 'roles optional' holds under that asymmetry.

Figure 3. The completed factorial and the cost–recall frontier.



(a) Top-3 recall of the four cells of the completed factorial on CPC: candidate lists from same-prompt sampling (A×5, temperature 0.7) or from five isolated specialist roles, integrated either by mechanical Borda aggregation or by the LLM moderator. Both moderator cells exceed both mechanical cells (which sit at or below the level of the single call, A×1 = 76.1%) and are statistically indistinguishable from each other, so the specialist framing of the lists is optional: the gain requires independent generation plus moderated synthesis. A×5 samples at temperature 0.7 because rank aggregation requires diverse samples — a design requirement of that arm, not a freely matched setting. Values are mean ± SD across five seeds.

(b) Top-3 recall versus tokens per case (log scale) for every CPC arm: A×1 (2,846 tokens), A×1+CoT (3,499), P (3,872), A×5 Borda/self-consistency (14,224), A×5+Mod (17,419) and MDT (19,422). The frontier runs A×1 ⟶ A×5+Mod ⟶ MDT: the +5.3-point top-3 gain of A×5+Mod over A×1 costs about six times A×1's tokens, and replacing the five plain samples with five specialist role calls buys a further 0.2 points. Judge: GLM-5.3-flash × v3; error bars ±1 SD across five seeds.

The benefit reverses in real emergency department cases

On ER-Reason — 364 real emergency department encounters whose inputs end before any laboratory or imaging results — the benefit of multi-perspective reasoning reversed in direction. A×1 outperformed MDT on every metric: top-1 40.1% versus 34.3% (−5.8 points, 95% CI −8.7 to −3.0, $p < 0.0001$), top-3 54.6% versus 45.5% (−9.0 points, $p < 0.0001$), and top-5 61.6% versus 51.9% (−9.8 points, $p < 0.0001$), with P intermediate (above MDT on all three endpoints; below A×1 on both recall endpoints and numerically at top-1, $p = 0.12$). The deficit was present and significant in both reference-label strata (symptom-level and disease-level), and a judge-independent check confirmed it: A×1 covered 38.3% of the reference label's content words in its top-5 lists against 30.7% for MDT ($p < 0.0001$), the same ordering in both strata (Figure 4b).

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Temperature does not explain the reversal: with all arms matched at 0.3, the ordering A×1 > P > MDT survived on every dataset (Table 1 note; Supplementary Note: Temperature-matched analyses).

Table 2. ER-Reason performance under the two input conditions (n = 364, primary judge, five seeds).

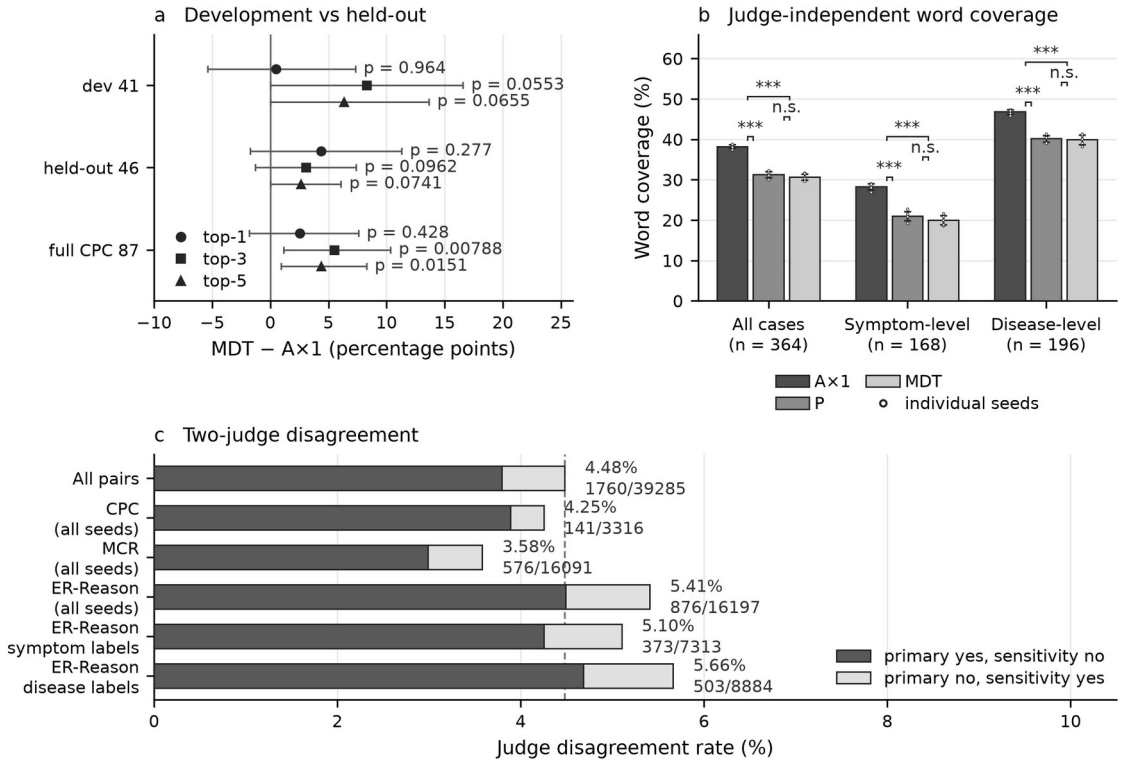
Strategy	Input	Top-1 (%)	Top-3 (%)	Top-5 (%)
A×1	Presentation only	40.1 ± 0.9	54.6 ± 1.1	61.6 ± 0.8
	Plus objective results	42.9 ± 0.8	57.2 ± 0.8	66.4 ± 0.6
P	Presentation only	37.7 ± 0.6	48.5 ± 1.5	55.2 ± 0.8
	Plus objective results	41.9 ± 0.8	53.7 ± 2.1	59.7 ± 1.9
MDT	Presentation only	34.3 ± 1.0	45.5 ± 1.0	51.9 ± 1.1
	Plus objective results	36.2 ± 0.5	48.7 ± 1.3	56.1 ± 1.3

Values are mean ± SD across five seeds; table hit rates are per-seed means, whereas the gains and interaction tests quoted here and in the text are case-level (per-case differences in five-seed hit rates), so the two conventions differ slightly on the ER workup cells (MDT top-5: 56.1% per-seed mean, 55.9% case level). Gains were +2.8 / +2.6 / +4.7 points (A×1), +4.2 / +5.2 / +4.5 (P), and +1.9 / +3.3 / +4.3 (MDT) at top-1 / top-3 / top-5. MDT under the objective-results condition excludes one case at top-3 and two at top-5 whose candidate–reference pairs could not be adjudicated after retries, counted as misses, the same convention as the P-split arm (Methods). Per-seed values for both conditions and the full interaction output appear in Supplementary Table S6.

Figure 4. Boundary conditions: prompt-development optimism, judge-independent word coverage, and two-judge disagreement.

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(a) Effect size of MDT against A×1 (percentage points) on the 41 development cases used for prompt design, the 46 held-out CPC cases never exposed to prompt tuning or judge refinement, and the full 87 CPC cases, scored by the primary judge only. Markers give case-level five-seed hit-rate differences, whiskers the 95% percentile bootstrap interval (10,000 case-level resamples, seed 20 260 917); p values are paired two-sided Wilcoxon on the same rates; all values in Supplementary Table S7. The recall-endpoint effect is roughly twice as large on the development cases as on the held-out cases, and no held-out contrast reaches significance at this sample size; the sensitivity judge agreed in direction on every held-out endpoint (top-1 +8.7 points, p = 0.064, not shown).

(b) Judge-independent check of whether the reference diagnosis is actually represented in the returned list, measured without any LLM judge, on all five seeds of ER-Reason. Content words are the tokens of the normalised reference label (lower-cased, non-letters replaced by spaces) longer than four characters; a case's coverage is the fraction of those words occurring as substrings of the normalised union of its top-5 candidates, and cases whose label has no content word (5 of 364) count as covered — they contribute identical (maximal) coverage to every strategy, can raise absolute levels but never a between-arm contrast, and enter the paired test as ties. Bars give mean coverage across seeds ± 1 SD, white dots individual seed values; strata follow reference-label granularity; brackets give the paired Wilcoxon test on per-case five-seed coverage, unadjusted. A×1 covered more of the reference label than either multi-perspective strategy in every stratum (all cases 38.3% versus 31.3% for P and 30.7% for MDT; both $p \le 1.2 \times 10^{-13}$), while P and MDT did not differ ($p \ge 0.13$).

(c) How often the two judges reach different verdicts on the same candidate–reference pair. The primary judge (GLM-5.3-flash) and the sensitivity judge (deepseek-v4.1-flash) ran under identical v3 rules, blinded to strategy; dark bars are pairs the primary judge accepted and the

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sensitivity judge rejected, light bars the reverse. Rows give all adjudicated pairs and the same quantity by dataset and, within ER-Reason, by reference-label granularity (fractions and denominators in Supplementary Table S8). Disagreement is high nowhere (3.6–5.7% in every stratum) but is most frequent where the reversal is largest (ER-Reason), which bounds how much of that reversal can be attributed to judge noise; the dashed line marks the overall rate.

The obvious competing explanation — that the presentation-only input is simply too thin — was tested by re-running the same 364 encounters with the objective results of each visit appended (laboratory values, vital signs, and imaging, electrocardiogram and ultrasound impressions, extracted by rule-based filters that never consult the reference diagnosis). Every strategy improved by broadly similar margins, the ordering was unchanged, and the strategy-by-condition interaction was null at every endpoint (MDT-versus-A×1: −0.9, +0.6, and −0.5 points at top-1/3/5, all $p \geq 0.57$; excluding the 34 cases with no objective content, or the 20 in which the reference label appeared literally in the added text, changed nothing) (Table 2; Supplementary Tables S6, S9). Input density is not what governs the reversal — an interaction larger than roughly 3.5 points is excluded at this resolution, not the possibility of any interaction (Supplementary Note S29).

The completed-factorial arm closes the attribution on this corpus as well: feeding the moderator five plain samples of the single-call prompt (Ax5Mod) erased the reversal entirely — within noise of A×1 at every endpoint (top-3 −0.8 points, $p = 0.39$) yet 8.2 points above the role-based team at top-3 ($p < 0.0001$) — and the moderator again beat mechanical aggregation of the same samples by +9.0 points at top-1. Multi-call inference per se is therefore not what harms on real emergency presentations — plain samples plus a moderator are exactly as safe as one call — the harm enters with the specialist role lists; the moderator's synthesis decides outcomes in both directions on both distributions (Supplementary Note S30). The same decoding caveat as in the CPC factorial applies here.

An exploratory, single-reviewer manual review of cases where A×1 was correct and MDT wrong suggests a mechanism (offered as a hypothesis, not as a coded causal claim): each isolated role amplifies a mechanistic or exotic hypothesis from the same thin evidence, and the moderator's synthesis promotes these elaborated hypotheses over the plain label that matches the emergency-level reference diagnosis. Where the reference label was ileus, all five roles converged on a motility-related syndrome and no obstruction-level diagnosis appeared in the top 5; further examples are given in Supplementary Note S28.

Failure taxonomy of the discordant pairs

Classifying the CPC top-1 discordant pairs (25 MDT wins, 20 A×1 wins, deduplicated; exploratory and descriptive, not hypothesis-tested) located the two strategies' failures on opposite axes: the dominant failure of A×1 was near-miss within-family discrimination (10 of 25 losses: the right family, defaulted to its most prototypical member), whereas every failure unique to MDT was zebra-related in one direction or the other (6 of 20: rare golds missed for common entities, or common golds escalated to exotic etiologies), and anchoring failures were bidirectional and concentrated in seven trap cases that flip across seeds (full taxonomy in Supplementary Note S31).

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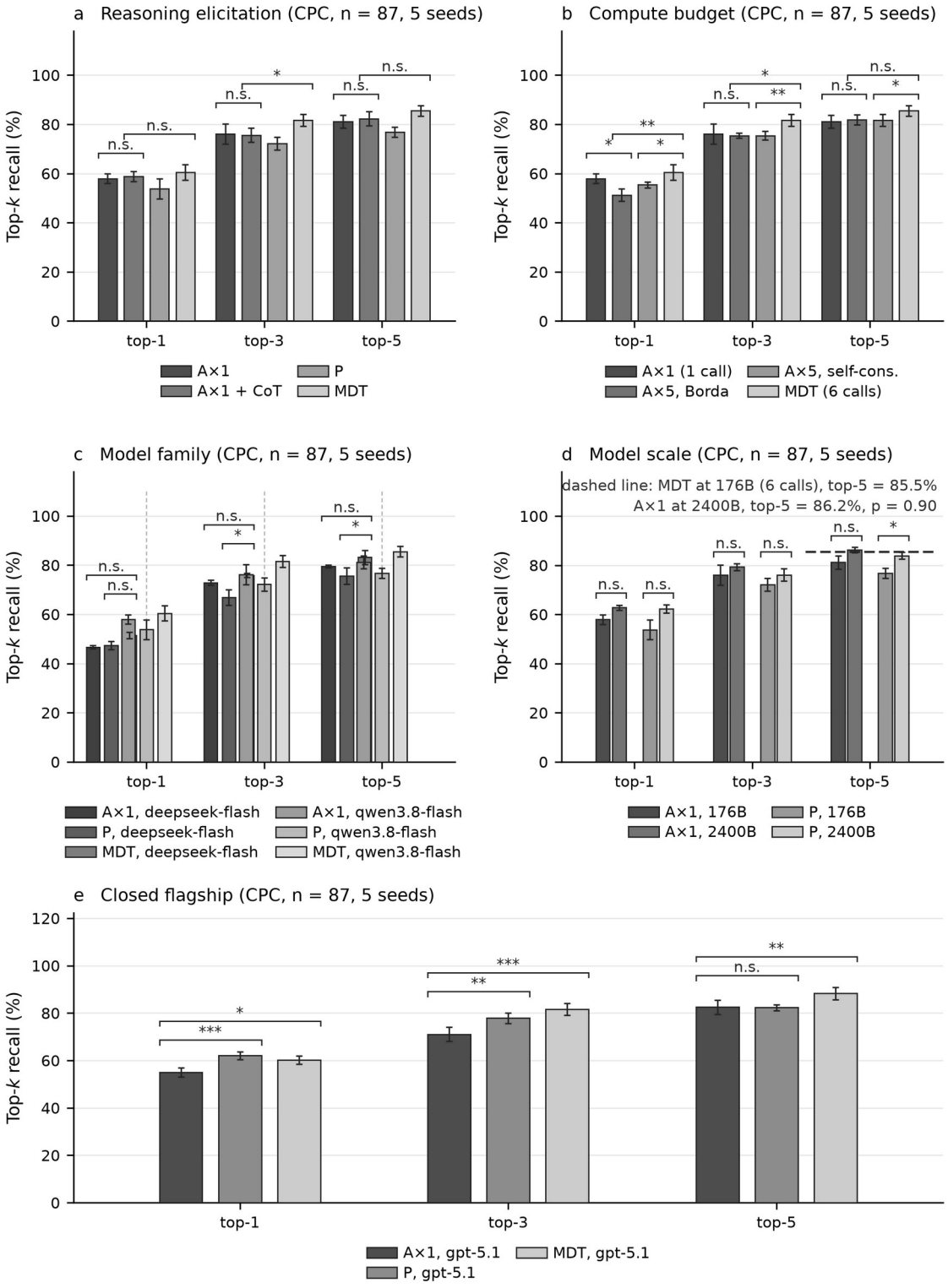
The pattern across model families and scales

All results above come from one open-weight backbone, so the isolation effect could in principle be family-specific. Re-running the three strategies on a second open-weight family (deepseek-v4.1-flash, byte-identical prompts, five seeds) reproduced the ordering: MDT ahead of P on both recall endpoints (+9.0 points at top-3, p = 0.017; +7.6 at top-5, p = 0.023) and ahead of the single call on point estimates at every endpoint (+4.8/+3.0/+3.7 points, none individually significant at n = 87), with P again below the single call (Figure 5c; Supplementary Table S11). On a closed flagship (gpt-5.1, five seeds) the persona effect reversed in sign — P rose +7.1 points above the single call at top-1 (p = 0.0009) and +6.9 at top-3 (p = 0.0070), where it had sat below or level with it on both open families — but stopped at the head of the list (top-5, p = 0.77), while the team effect showed no such dependence: MDT separated from the single call at every endpoint including top-1 (top-3 +10.9 points, p = 0.0003; top-5 +6.0, p = 0.005; top-1 +5.6, p = 0.016) (Figure 5e; Supplementary Table S17). The persona effect is thus family-dependent in sign, and the team's recall advantage is not: three families, three positive top-5 margins. The main-text P-arm numbers come from the original prompt, which carried a transcription defect (Methods); the corrected-prompt control raises P recall by 3–4 points on CPC without changing any persona conclusion, and both versions are reported (Table 1; Supplementary Table S10c).

Figure 5. Controls that isolate the source of the benefit: reasoning elicitation, compute budget, model family, model scale, and a closed flagship.

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All five panels use the CPC corpus (n = 87), the same frozen judge (GLM-5.3-flash, v3 rules, blinded to strategy), and are directly comparable with the CPC column of Figure 2. Bars give mean top-1, top-3 and top-5 recall and error bars ± 1 SD across seeds (five seeds in every panel). Within each endpoint, bars follow the panel-legend order, each in its own grayscale

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tone. Brackets give the two-sided paired Wilcoxon test on per-case hit rates, with *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant; all p unadjusted, these controls lying outside the pre-specified 27-test correction family. Full statistics are tabulated in Supplementary Tables S10–S12 and S17.

(a) Reasoning elicitation. A×1+CoT is the single call asked for a short step-by-step justification before the ranked list (identical prompt skeleton, guidelines, JSON schema, model and temperature; the elicitation trace appeared in all 435 calls, mean 2,713 characters). Eliciting that reasoning changed nothing against A×1 (all $p \geq 0.61$), while MDT stayed ahead at top-3 ($p = 0.027$) and directionally ahead at top-5 ($p = 0.067$), so the team's advantage is not explained by the presence of intermediate reasoning text. P and MDT are shown for reference.

(b) Compute budget. A×5 spends five calls per case on the single-call prompt, aggregated by Borda count over clustered disease labels or by self-consistency majority voting on the rank-1 cluster; MDT spends six calls structured as isolated roles. Neither aggregation reproduced the team: Borda fell below A×1 at top-1 ($p = 0.017$), both tracked A×1 on recall, and both sat below MDT at every endpoint ($p \leq 0.059$ at top-5). Extra test-time compute therefore buys recall without reproducing the isolated-role advantage.

(c) Model family. The three strategies were re-run end to end on deepseek-v4.1-flash (five seeds, byte-identical prompts) and scored by the same judge; the qwen3.8-flash arms are the main-experiment runs over the same five seeds. The ordering reproduced in the second family — MDT ahead of the single call on point estimates at every endpoint (top-1 +4.8 points, $p = 0.067$) and ahead of the persona arm on both recall endpoints (top-3 $p = 0.017$, top-5 $p = 0.023$), with P never above the single call — so the structure effect is not specific to one model family, although only the qwen MDT-versus-A×1 contrasts are individually significant and absolute levels differ between families. Panel (e) extends the same comparison to a closed flagship, gpt-5.1 (five seeds; also tabulated in Supplementary Table S17).

(d) Model scale. Scaling the backbone from 176B to 2400B parameters raised every endpoint of both single-call strategies, with no reliable strategy difference in that gain (interaction $p \geq 0.43$). The six-call 176B team (dashed line, 85.5 ± 2.1% at top-5) was statistically indistinguishable from the single 2400B call (86.2 ± 1.1%, $p = 0.90$) — indistinguishability at these sample sizes, not demonstrated equality.

(e) Closed flagship. The three strategies re-run end to end on gpt-5.1 (five seeds, byte-identical prompts, same frozen judge). The persona effect reversed in sign — P rose +7.1 points above the single call at top-1 ($p = 0.0009$) — without extending the differential (top-5, $p = 0.77$), while the team separated from the single call at every endpoint (top-3 +10.9 points, $p = 0.0003$; top-5 +6.0 points, $p = 0.005$; top-1 +5.6 points, $p = 0.016$). Full statistics in Supplementary Table S17.

Heterogeneous roles and inference cost

Replacing two of the five isolated roles with a different model family left performance essentially unchanged (all pairwise $p \geq 0.90$ against the homogeneous team over three seeds, underpowered for equivalence), consistent with the advantage stemming from the isolation-and-synthesis rather than the identity of the weights (Supplementary Note S32). The strategies

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differ sharply in cost: MDT consumed 6.8–8.4 times the tokens of a single call across the three datasets (the emergency figure is a single-seed measurement; Table 3), and the factorial simplifies the recipe — plain repeated samples plus one integrating call achieve the recall gain at slightly lower cost than the role-based team (17,419 versus 19,422 tokens; Figure 3b). The moderator also adds a serial synthesis step to wall-clock latency, since it waits for all five role calls. That premium buys recall on the datasets whose reference labels reward discriminating among near neighbors, and buys nothing — indeed converts to harm — on the emergency encounters.

Table 3. Token usage per case.

Dataset	Strategy	Calls per case	Total tokens per case (mean ± SD)
CPC (n = 87)	A×1	1	2,846 ± 1,268
	P	1	3,872 ± 1,302
	MDT	6	19,422 ± 7,686
	A×1+CoT	1	3,499 ± 1,421
	A×5	5	14,224 ± 6,333
	A×5+Mod	6	17,419 ± 7,650
MCR (n = 406)	A×1	1	850 ± 123
	P	1	1,797 ± 198
	MDT	6	7,134 ± 864
ER-Reason — single seed (1 × 364); SDs are case-to-case, not comparable with five-seed rows			
ER-Reason (n = 364)	A×1	1	1,686 ± 473
	P	1	2,675 ± 505
	MDT	6	11,936 ± 2,914
CPC (n = 87, deepseek-v4.1-flash) — five seeds			
CPC (n = 87, deepseek-v4.1-flash)	A×1	1	2,591 ± 1,148
	P	1	3,739 ± 1,157
	MDT	6	17,926 ± 6,960

Token counts are per-example totals recorded at inference; the logs do not separate input from output tokens, so we report token volume rather than imputed currency costs. The CPC control rows cover the same five seeds as A×1, P, and MDT; A×5 sums its five samples per case

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(matching MDT's call count but not its token total), and the A×5+Mod row adds the moderator call to those samples (17,419 = 14,224 + 3,195); the A×5 and A×1+CoT rows are recomputed from their per-case run files and deposited with the per-seed breakdown in routing_study/results/token_costs/control_arms_tokens.json. The ER-Reason rows come from a single seed (1 × 364) and their standard deviations describe case-to-case variation only, whereas the CPC and MCR rows aggregate five seeds (5 × 87, 5 × 406); the deepseek-v4.1-flash rows cover CPC only, and the prompt-corrected P arm uses the same prompt and provider as P, so it carries no separate cost row (Supplementary Table S10c). Record coverage was 100% across all strategy × dataset × seed combinations reported here; the matched control arms are collected in Supplementary Tables S10–S12.

Discussion

Across three datasets differing in difficulty, provenance, and reference-label structure, the benefit of multi-perspective LLM reasoning is real but conditional, with a component-level attribution. The active pair is independent generation of the lists combined with moderated synthesis over them: mechanical aggregation of the same lists fails from either source, the moderator over conditioned voices fails, and either list source recovers the full gain under the moderator — so the specialist personas are optional, on every corpus. The benefit concentrates in differential-list recall rather than top-1 accuracy — the team's top-1 margins on the open-weight families were directional only ($p = 0.43$, $p = 0.067$), the exception being the strongest backbone tested (+5.6 points, $p = 0.016$) — and it reverses outright on real emergency department presentations, where the reference diagnosis is the treating clinician's working label. On the curated corpora, the judge-independent word-coverage check reproduced the persona penalty but not the team's lexical advantage: A×1 matched (CPC, +2.3 points, $p = 0.062$) or slightly exceeded (MCR, +1.7 points, $p = 0.019$) MDT in literal coverage, so the recall gain is carried by semantically equivalent rephrasings accepted by the clinician-validated judge — the same evidentiary basis as every adjudicated LLM diagnostic benchmark (Supplementary Note S34).

The failure analysis supplies a mechanism for this boundary. Where the team won, it won by discriminating among near neighbors within a disease family — the single call's most common failure — and where it lost, it lost on zebras in both directions, missing rare diagnoses and inventing exotic ones. Moderated re-adjudication over independently generated candidates sharpens within-family discrimination; the same integrative freedom, applied past the evidence, licenses etiologic overreach. The emergency cases are the extreme of this trade: when the reference label is the clinician's own working diagnosis, elaboration is pure loss — and remained pure loss when the visit's objective results were added.

These conditions reconcile a contradictory literature. The positive reports employed multi-call designs in which contributions are generated in separate steps and then aggregated [8,13]; the negative reports voiced personas within a single prompt, where perspectives condition on one another and collapse toward a consensus [14], or debated without external feedback [15]. Our P condition reproduces the negative result and our MDT condition the positive one under a single backbone, with isolation as the only manipulated variable — an attribution the confounded literature could not make. The same reading explains why automatically assembled multi-agent systems underperform budget-matched baselines while hand-crafted expert designs do not [17], why collaboration's benefit shrinks as capability grows [22], why single

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agents match multi-agent systems at equalized thinking budgets [23], and why tool-using clinical agents struggle to beat plain baselines [24]: structure, not agent count, carries the effect. The mixed-vendor concern raised against conversational frameworks [18] finds no support here either — swapping two roles to a different model family changed no metric — although at this sample size that is absence of evidence, not evidence of absence.

Three matched alternatives each fail against a control: intermediate reasoning text inside the single call moved nothing ($p \geq 0.61$); a five-fold call budget with mechanical aggregation bought the single call's recall, never the team's; and the particular model family cannot explain the recall effect, which is positive in all three families tested, while the persona effect flips sign with the family — helping the strongest backbone at the head of the list (top-1 +7.1 points, $p = 0.0009$; top-3 +6.9, $p = 0.0070$) yet adding nothing to the breadth of the differential (top-5, $p = 0.77$).

One asymmetry orders the deployment reading: the attribution is internal to each corpus, while the only corpus whose labels no model could have memorized is the emergency set, where the method reverses.

The results translate into deployment guidance keyed to the clinical question, the input at hand, and now the backbone. When the task is a single most-likely diagnosis, the single direct call remains the right default on the open-weight families — nothing beats it robustly at top-1, at six times fewer calls and an order of magnitude fewer tokens — but the default is not a law: on the strongest backbone tested, both multi-perspective arms beat the single call at top-1, so the case for paying the multi-call cost strengthens with capability, consistent with the capability-axis analysis of Kim et al. [22].

What does not change with the backbone is the breadth prescription: when the clinical value lies in the differential's breadth — a rare-disease service assembling an initial work-up framework, a referral decision where missing the second-most-likely diagnosis costs more than list length, a teaching ward generating starter differentials for trainees — an integrating call over independently generated candidate lists justifies its token premium on both open-weight families. When the reference label is the treating clinician's working diagnosis, as in first-presentation emergency care, role-based multi-perspective prompting should be avoided under a working-label-matching criterion, the only criterion we could benchmark; the plain-sample moderator recipe is the defensible multi-call design there, if any multi-call design is used at all.

One circularity qualifies every emergency-side recommendation: symptom-level working-diagnosis labels are intrinsically closer to the plain, presentation-matched answer a single call produces, so a method that elaborates toward disease-level entities starts at a granularity disadvantage under a word-level equivalence criterion. The reversal, however, is not a granularity artifact — it is significant within both label strata — but when the clinical question is a differential list and the quality criterion is a working diagnosis, the fairness of any top-k metric is itself unresolved. Strategy selection, in other words, behaves like a routing problem whose variables are observable before the model is run; we offer this as a hypothesis for prospective evaluation, not as a validated deployment rule.

Several limitations qualify these findings. The primary emergency input stops before the Medical Decision Making section, which structurally weakens the imaging and laboratory role; the workup-informed condition added objective results only, not the clinician's assessment and

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plan, so the emergency findings should not be extended to multi-agent reasoning on a complete chart; its stratified tests are exploratory and uncorrected for multiplicity. The primary judge belongs to a different family from the backbone it scores, which cuts both ways: the cross-family judge-judge disagreement bounds, but does not eliminate, judge-induced error; the larger gpt-5.1 contrasts exceed that bound, but the narrower nulls claim no more than absence of evidence; and no blinded clinician rated the discordant emergency pairs for acceptability, so the reversal is established at the level of label matching, not of clinical judgement. Because NEJM CPC cases are published, a stronger backbone may have memorized them; the emergency corpus, whose labels no model could have memorized, shows the opposite ordering, but we cannot exclude that memorization interacts with strategy. The moderator-anatomy ablations are exploratory and single-seed; the heterogeneous-role comparison is underpowered for equivalence; the backbone-scale comparison cannot resolve the interaction it reports as null; and the deployment guidance itself awaits prospective evaluation against outcomes that matter clinically rather than benchmark labels. Nor was the analysis externally registered: pre-specification rests on the dated, append-only milestones of the judgment cache and the locked evidence bundle rather than on a registry protocol, and several control arms — temperature matching, the completed factorial, the corrected-prompt control — were added post hoc and are labelled sensitivity analyses wherever they bear. Finally, the evidence is asymmetric in a deployment-relevant way: the positive gains come from curated corpora that stronger backbones may have seen, whereas the one corpus no model could have memorized — real first-contact encounters — shows harm; gains measured on curated benchmarks should not be extrapolated to front-line inputs, and on the curated corpora the team's recall advantage is invisible to a literal word-coverage criterion, resting on judge-adjudicated semantic equivalence (Supplementary Note S34).

Methods

Study design and datasets

We compared three diagnostic prompting strategies on three case datasets that differ in difficulty, provenance, and reference-label structure; all strategies received the same case text and returned a ranked list of five differential diagnoses, scored against the reference diagnosis of each case. The development corpus was 87 clinicopathological conference cases published consecutively as Case Records of the Massachusetts General Hospital over a fixed publication window (2013–2016) and curated in the benchmark of [5], split into 41 used for prompt design and judge refinement and 46 held out from prompt tuning and model selection. For external validation we used 406 MedCaseReasoning cases [20]; released in 2025, its cases may have entered the training data of the evaluated models, a contamination risk we acknowledge. The third dataset, ER-Reason, contains real emergency department encounters from a PhysioNet credentialed repository used under its data use agreement [21]; from 3,984 encounters we drew a stratified random sample of 364 after patient-level deduplication, by chief complaint under a fixed sampling seed (procedure and record-coverage detail in Supplementary Note: ER-Reason sampling). The model input was the ED provider note truncated immediately before the "Medical Decision Making" section (fallback markers where needed; no encounter lost to missing text); the HP_Note_Text column does not align with its encounter, so ED_Provider_Notes_Text was used throughout. Reference labels exist at two granularities, symptom-level (n = 168) and disease-level (n = 196), a mixture inherent to the source record.

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All API calls receiving these encounters' text followed the zero-data-retention OpenRouter routing described under Models and decoding; adjudication, which involves only reference labels and candidate diagnosis strings, ran on the primary-judge endpoint.

Workup-informed input condition

To test whether the emergency reversal reflects the thinness of presentation-only input, we built a second input for the same 364 encounters: the presentation text is reused byte-for-byte and the objective results of that encounter are appended, with prompts, model, decoding, seeds, and judge unchanged. Rule-based filters that never consult the reference diagnosis retain laboratory values, vital signs, point-of-care results, and imaging, electrocardiogram, and ultrasound impressions, and remove assessment and plan statements, dispositions, handoffs, prescriptions, and summary sentences; manual review of every remaining assertional sentence found no explicit final diagnosis asserted in the added text. Coverage is bounded by the source record: 240 of 364 cases (65.9%) have at least 200 characters of objective content and 34 have none, for which the two conditions are identical by construction. A content-word flag identified 20 of 364 cases (5.5%) in which the added text effectively contains the reference label (8 by a stricter phrase rule), and the sensitivity analysis excludes the flagged cases.

Prompting strategies

All strategy calls used qwen3.8-flash (176B parameters, provider-documented) with thinking mode disabled; all strategies produced the same output format, a ranked top-5 list. A×1 is a single zero-shot call that returns the list directly, eliciting no intermediate reasoning text, at requested temperature 0. P reasons sequentially from five perspectives inside one shared context — attending physician (unifying diagnosis), pathophysiology, imaging and laboratory, epidemiology and exposure, and a skeptic — at temperature 0.3, in one call. MDT issues the same five roles as independent calls that cannot see one another's output, and a moderator call integrates the five opinions, for six calls per case. P and MDT share roles, model, and prompt material; the only manipulated variable is whether the perspectives are isolated. Full prompt text is given in Supplementary Methods S-M1–S-M3.

Matched control arms

Beyond A×5 (five independent samples of the single-call prompt at temperature 0.7, aggregated by Borda count or self-consistency majority vote), the control arms comprise A×1+CoT (reasoning elicitation); MDT-Borda (the team's role lists re-aggregated without the moderator); Ax5Mod (the same moderator over plain samples, with fresh-sampled ER-Reason and MedCaseReasoning counterparts); P-split (sequential perspective analyses integrated by a fresh moderator); single-seed moderator-anatomy ablations; the second model family (deepseek-v4.1-flash, byte-identical prompts, five seeds, including the emergency factorial); a closed-flagship extension (gpt-5.1, five seeds); the prompt-corrected Pfixed; the 2400B backbone-scale arm; and a heterogeneous-role team. Per-arm scripts, prompts, and deposited run files are listed in Supplementary Note S33 and Supplementary Tables S10–S12 and S17.

Data integrity repair

Two defects found after the main runs were repaired before analysis: 16 empty role-call outputs in the MedCaseReasoning MDT arm were re-run, the affected case-seeds re-

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synthesized and re-adjudicated under the frozen judge, and five empty rows in the 2400B P arm were re-run. Pre-repair files are retained as .bak_prefill backups beside the repaired ones and the repairs are logged (routing_study/results/refill_defects.log); the repaired MCR numbers moved by at most 0.2 points and no conclusion depends on the repair (Supplementary Note S14).

Models and decoding

Models are identified by the deployment strings of their serving endpoints: qwen3.8-flash via DashScope, deepseek-v4.1-flash (API alias deepseek-flash) via the DeepSeek API, GLM-5.3-flash via the Zhipu API (321B, primary judge), a 552B deepseek-v4.1-flash sensitivity judge, and gpt-5.1-2025-11-13 via an OpenAI-compatible endpoint. Parameter counts are as documented by the providers, which return no immutable snapshot hashes; runs are identified by the deposited request/response logs. All arms ran between 2026-09-10 and 2026-09-23. All API calls that receive ER-Reason case text — strategy generation and any synthesis over case content — were routed through OpenRouter (openrouter.ai) with zero-data-retention routing enabled (provider zdr: true, restricting traffic to endpoints that retain no request or response data; endpoint data collection disabled, data_collection: deny). Judge adjudication exchanges only the reference label and a candidate diagnosis string, no case content, and ran on the primary-judge endpoint. Requested temperature 0 is not guaranteed greedy decoding: an illustrative three-case probe resubmitting identical prompts changed the top-1 diagnosis in one case, so each "seed" is an independent repeat measuring run-to-run variance, and the case, not the run, is the unit of inference. A×1 requested temperature 0, the multi-perspective arms 0.3; because that asymmetry confounds temperature with strategy, all three arms were re-run at temperature 0.3 on every dataset (Results; Supplementary Note: Temperature-matched analyses).

Prompt transcription defect and its control

The P strategy appends a fixed output-format suffix that, unlike the A×1 prompt, was never passed through Python's .format(), so its JSON example reached the model as literal doubled braces. qwen3.8-flash rewrote the braces itself (none of the 6,105 qwen-side P outputs was empty or carried fewer than five candidates), but deepseek-v4.1-flash copied them verbatim: 57 of the 261 rows of seeds 1–3 (21.8%) returned an unparseable list on the first attempt, all recovered — 49 on retry and 8 via a parser-level fold — and 56 of the 435 final rows (12.9%) passed through the flagged fold in total. The byte-exact text actually sent, with SHA-256 fingerprints of each prompt constant, is deposited as routing_study/results/prompt_actually_sent.md. As a diagnostic for fold bias, the 56 rows that passed through it do not differ systematically in adjudicated accuracy from the 379 natively parsed rows: the per-case paired comparison over the 39 cases containing both kinds of rows is balanced at top-1 and top-5 (sign test $p = 1.0$; the only marginal stratum contrast, top-5 within seeds 4–5, $p = 0.046$, does not survive case-level pairing) — if anything the fold is conservative for the persona arm, as repaired rows echoed the defect. Re-running P with the braces folded (Pfixed; differing from P on that one line only, audited case by case in the producing script) raises P recall by 3–4 points on CPC, leaving the team's recall advantage over the persona no longer individually significant there; the corrected arm still failed to beat the single call at any endpoint, and on the external set the persona conclusions are unchanged. Both versions are reported (Table 1; Supplementary Table S10c).

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LLM judge and validation

Because correct diagnoses surface under synonyms, abbreviations, and differing specificity, scoring used an LLM judge deciding semantic equivalence between each candidate and the reference diagnosis; judgments were cached, one verdict per candidate–reference pair; the primary judge (GLM-5.3-flash) and rule version were frozen before the outcome analyses; the cache stands at 113,303 pairs. Rule development used a boundary-enriched batch reported only in Supplement (Supplementary Table S5); validation used a disjoint 100-pair batch rated by two blinded senior clinicians who are not co-authors of this study (a chief physician of respiratory medicine and an associate chief physician of hematology) and was extended to MedCaseReasoning and ER-Reason with the same protocol, including a voided first-pass criterion on ER-Reason that was re-done under an explicit main-diagnosis test (Supplementary Table S5-2); agreement is reported in Results. A second judge (deepseek-v4.1-flash) ran under identical rules as sensitivity judge on every run of all three datasets (disagreement 4.48% overall, 3.6–5.7% by stratum; Supplementary Table S8). The released caches exclude entries derived from ER-Reason under its data use agreement.

Statistical analysis

The pre-specified primary endpoints were top-3 and top-5 recall; top-1 accuracy is reported for completeness but carries no superiority claim. The endpoints, the primary judge, and the two comparison families (MDT versus A×1; MDT versus P) were fixed before the strategy comparisons were run; with no external registration, pre-specification is documented by the dated, append-only growth milestones of the judgment cache and the locked evidence bundle. All datasets were run under five seeds; strategies are compared across cases with the paired two-sided Wilcoxon signed-rank test on per-case five-seed hit rates, with 95% percentile confidence intervals from a case-level cluster bootstrap (10,000 resamples) and, secondarily, an exact McNemar test on majority-vote outcomes; threshold-sensitivity at t = 2–5 of five seeds and a case-clustered GEE re-fit of all 27 primary tests converged and confirmed every conclusion (Supplementary Tables S15, S16) — three frameworks that also answer, in advance, the heavy-tie concern about the Wilcoxon approximation on discrete five-seed hit rates — and pooling case–seed pairs as independent is retained only as a legacy reference (Supplementary Table S3). We report unadjusted p values; all main conclusions survive Benjamini–Hochberg correction across the 27 tests of the primary framework (the sole exception, ER-Reason MDT-versus-P top-3, q = 0.054, carries no conclusion; Supplementary Table S4), and the four factorial contrasts survive correction within their family. Results are reported as mean ± SD across seeds. Power: the 87-case CPC corpus resolves 3–4-point recall differences but not top-1 differences below about 8 points, so its top-1 nulls are inconclusive by construction; the 364-case emergency sample resolves roughly 2–3-point top-1 contrasts; the CPC ceiling is set by case availability under the journal's copyright terms.

Data availability

CPC case text is copyrighted by the publisher (Case Records of the Massachusetts General Hospital, NEJM) and cannot be redistributed; we release case identifiers with pointers to the source journal and to the benchmark of [5]. MedCaseReasoning is publicly available under a CC-BY license [20]. ER-Reason is a PhysioNet credentialed dataset available to users who complete credentialing and sign its data use agreement [21]. All API calls receiving ER-Reason

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case text (strategy generation and synthesis) were routed through OpenRouter's zero-data-retention tier (Methods); judge adjudication, which exchanges only reference labels and candidate diagnosis strings, ran on the primary-judge endpoint. Per-case ER-Reason outputs and the ER-Reason entries of the judge caches are excluded from the public archive under the same agreement.

Code availability

All analysis code, prompt templates, pipeline scripts, and the judgment caches are openly released at https://github.com/yepsun/multi-agent-diagnostic-reasoning and permanently archived on Zenodo at https://doi.org/10.5281/zenodo.22935 060 (snapshot v1.0.1). The caches contain every adjudication used for CPC and MedCaseReasoning; entries derived from ER-Reason are excluded, and case identifiers are one-way hashed, in line with the ER-Reason data use agreement.

Ethics declarations

This study is a secondary analysis of existing, de-identified data and did not involve human subjects recruitment or access to identifiable information; institutional review board review was therefore not required. The CPC cases are published case reports; MedCaseReasoning is derived from open-access case reports in PubMed Central [20]. ER-Reason consists of clinical notes de-identified by the UCSF Information Commons under both HIPAA Safe Harbor and Expert Determination methods, with sharing approved by the source institution's compliance team, and was used in accordance with the PhysioNet data use agreement [21]. Consistent with PhysioNet's guidance on the use of credentialed data with large language models, all automated processing of these de-identified records ran on inference endpoints configured for zero data retention — no request or response retained by the provider, no use of the data for model training, no human review of case content — with every call receiving case text routed through OpenRouter's zero-data-retention tier (Methods and Data availability).

Author contributions

Jun Feng: conceptualization, methodology, writing — original draft. Yuhao Jiao: conceptualization, methodology, writing — original draft. Yiyao Li: methodology, software, formal analysis, investigation, visualization, writing — original draft. Longxin Xie: data curation, investigation, validation. Wenyu Peng: data curation, investigation, validation. Xuefeng Sun: conceptualization, supervision, funding acquisition, writing — review and editing. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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Declaration of AI use

Large language models were used to assist with drafting and language editing of this manuscript. The authors reviewed, verified, and take full responsibility for all content, including every reported number and citation.

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